

Optimal Electrode Design for Integrated Optics Modulators

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Abstract—We consider the problem of optimizing parallel stripe electrodes intended for influencing the refractive index of integrated optics waveguides that are embedded in electrooptic materials. In particular, such electrodes may be used for modulating the coupling efficiency of push-pull directional couplers or the phase of one of two orthogonal modes of a single waveguide. We show that an optimum electrode spacing exists for a push-pull directional coupler with fixed waveguide spacing. In addition, we discuss the optimum position for a single dielectric waveguide placed underneath a pair of stripe electrodes and show that it depends on the orientation of the optic axis of the electrooptic crystal and on the spacing of the electrodes.

INTRODUCTION

INTEGRATED optics employs modulators or switches that consist of one or two diffused dielectric waveguides that are embedded in an electrooptic material [1]–[4]. Switching or modulation may be accomplished by modifying the propagation constant of a guided mode by changing the refractive index of the electrooptic material, and hence of the waveguide, by an externally applied electric field that may be generated by two parallel metal stripes placed side by side on the surface of the substrate. Since the efficiency of the modulator or switch depends on the position of the electrodes and the waveguides relative to each other, we consider in this paper the problem of optimizing the electrode spacing and the position of the waveguide relative to the electrodes for two types of modulators: push-pull directional couplers and single waveguide devices [1]–[4].

Optical directional couplers can be used as switches or light modulators if the degree of coupling between the two waveguides can be altered by an externally applied electrical signal. Fig. 1(a) shows the waveguide and electrode configuration of such a device. The waveguides are usually defined by diffusion of a suitable dopant material into an electrooptic material such as lithium niobate. The application of an electrical field alters the index of refraction of the electrooptic material and hence, changes the propagation constant of a dielectric optical waveguide embedded in it. In an anisotropic material such as lithium niobate, the change of the refractive index depends on the direction of the applied electric field relative to the direction of the optical axis (c -axis) of the uniaxial crystal. In lithium niobate (and similar crystals) the refractive index change in the direction of the applied field is largest if the c -axis and field are parallel and the index change reverses sign as the

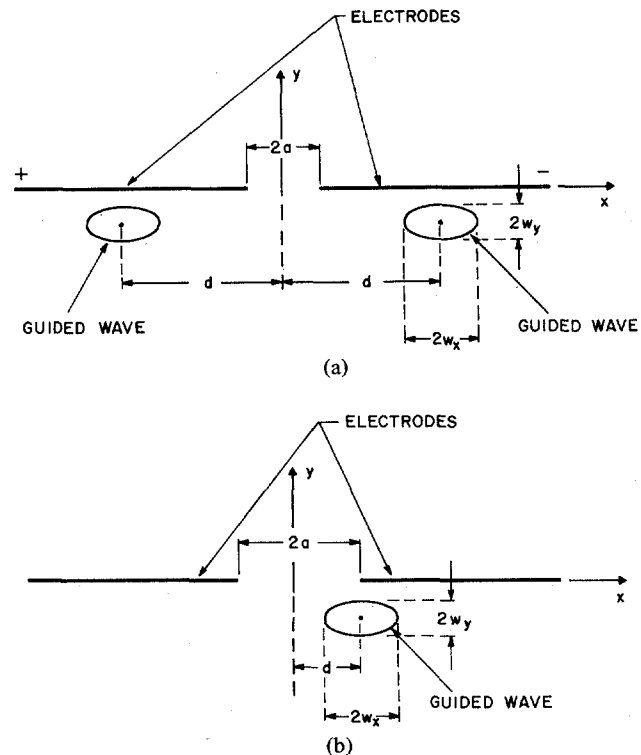


Fig. 1. (a) Schematic of "tunable" directional coupler. Metal electrodes and an outline of the mode fields guided by two diffused waveguides are shown. (b) Schematic of metal electrodes above a single dielectric waveguide.

direction of the electric field is reversed. Thus, if a voltage is applied to the electrodes shown in Fig. 1(a) the refractive index of the material can be enhanced under one electrode and decreased under the other so that push-pull "tuning" of the directional coupler can be accomplished.

It is natural to ask if there is an optimum spacing that maximizes the "tuning" effect of the directional coupler for a constant value of the externally applied signal voltage and fixed waveguide spacing. Since the electric field is strongest near the edge of the electrode, one may expect the strongest effect when the edge of each electrode is centered above each waveguide. On the other hand, as the electrodes are further removed from each other, the electric field intensity decreases for fixed voltage. The situation may thus not be quite so simple.

We shall show that for closely spaced waveguides, the strongest effect occurs when the electrodes are placed as closely as possible to each other. But when the optical waveguides are separated farther, an optimum electrode spacing exists that

Manuscript received August 31, 1981.

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favors placing the edges of the electrodes nearer but not directly above the centers of the waveguides. However, the maximum is relatively low so that for small waveguide separation little is lost if the electrodes are spaced as closely as possible.

The second problem to be discussed involves the question of the optimum location of a single waveguide under a pair of parallel stripe electrodes as shown in Fig. 1(b). This configuration is relevant if the relative phase velocities of two orthogonally polarized modes (TE and TM modes) are to be changed by an externally applied electric field. The answer to the problem depends on the way the crystal is cut. If the c -axis is perpendicular to the crystal surface, then the normal electric field component (E_y) is effective in changing the n_y component of the refractive index tensor. If the c -axis is parallel to the surface, then the parallel electric field component (E_x) affects the n_x component of the index tensor. The question of optimum waveguide position depends on these considerations and on the electrode spacing. If use is to be made of the parallel electric field components (E_x parallel to the crystal surface) it is best to locate the waveguide centrally between the two electrodes. Only for large electrode spacing is a slight advantage to be gained by spacing the waveguide near the edge of one electrode. If use is to be made of the perpendicular field component (E_y), which vanishes in the midplane between the two electrodes, the optimum waveguide position is found near the edge of one of the electrodes.

SHAPE OF THE dc ELECTRIC FIELD

For simplicity, we neglect the dc anisotropy of the electro-optic material and treat it as though it were isotropic. Even though the anisotropy will change the field configuration slightly, we do not expect that the result obtained by assuming an isotropic material will differ substantially from the corresponding result for an anisotropic material.

The field produced by the electrodes shown in Fig. 1 is described by the equations [5]

$$E_x = \frac{V}{\pi} \frac{\cos(\phi/2)}{[(a^2 + y^2 - x^2)^2 + (2xy)^2]^{1/4}} \quad (1)$$

$$E_y = \frac{V}{\pi} \frac{\sin(\phi/2)}{[(a^2 + y^2 - x^2)^2 + (2xy)^2]^{1/4}} \quad (2)$$

with

$$\tan \phi = \frac{2xy}{a^2 + y^2 - x^2} \quad (3)$$

V is the externally applied signal voltage and $2a$ is the spacing between the electrodes.

The direction of the electric field lines is shown in Fig. 2. In the gap at $y = 0$ the y component vanishes $E_y = 0$, while on the surface of the electrodes we have $E_x = 0$.

The distributions of the E_y and E_x field components as functions of x/a are shown in Figs. 3-7 at several (normalized) distances y/a from the electrodes. Fig. 3 shows that the field peaks sharply near the electrode edge at $x = a$. It also shows that $E_y = 0$ inside the gap for $x < a$ and that $E_x = 0$ on the surface of the electrodes at $x > a$. At greater distances from the electrodes the field maximum is no longer as large (note the different scales for different figures) and the fields no longer vanish at $x < a$ or $x > a$.

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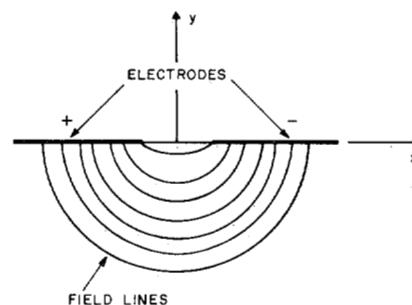


Fig. 2. Direction of the electric field lines produced by a dc potential applied to the two metal electrodes.

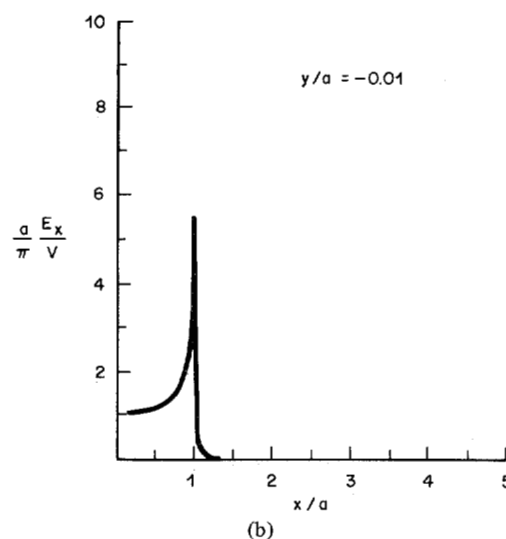
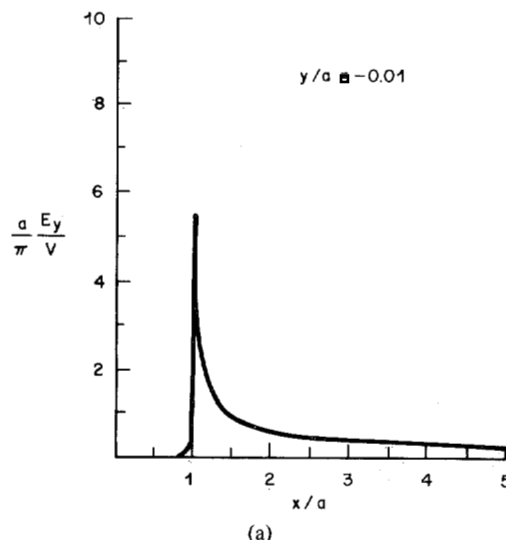
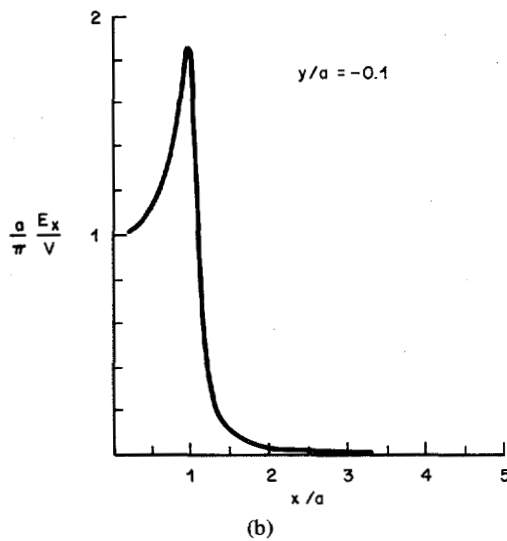
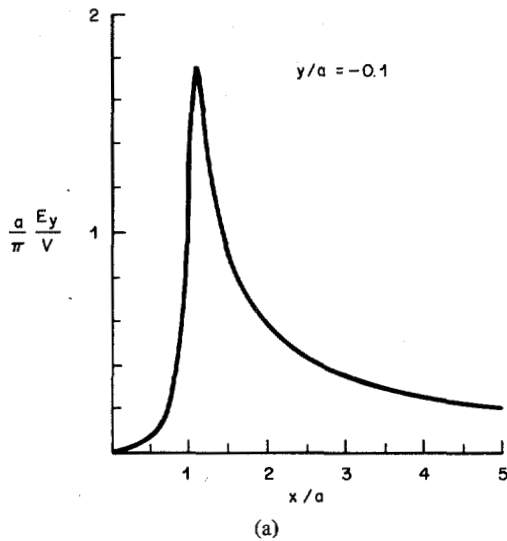
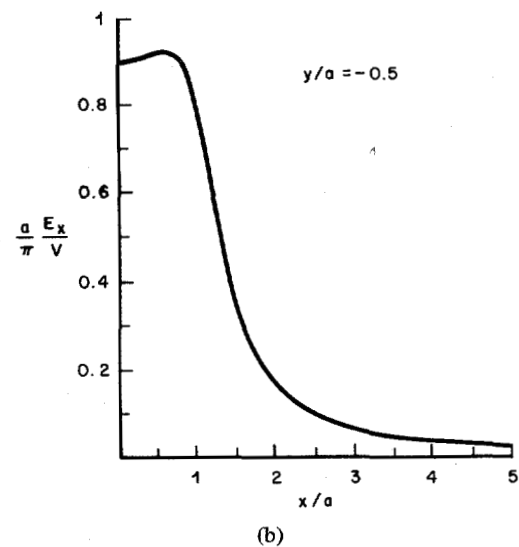
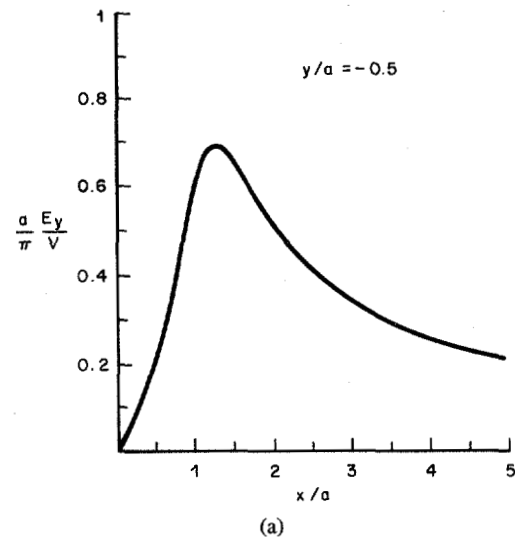


Fig. 3. Normalized electrical field components as functions of the normalized horizontal coordinate x/a at a horizontal line at distance $y = -0.01a$ below the surface of the electrodes: (a) shows the y component, and (b) the x component of the electric field.

We assume that the electric field vectors of the optical guided waves are polarized in the direction of the c -axis of the uniaxial electro-optic crystal. In case of a push-pull directional coupler the c -axis would be oriented perpendicular to the crystal surface, in other types of modulators the c -axis may be either perpendicular or parallel to the surface but always perpendicular to the waveguide axis. In any case, we are interested in the change of the refractive index in the direction of

Fig. 4. Same as Fig. 3 with $y/a = -0.1$.Fig. 5. Same as Fig. 3 with $y/a = -0.5$.

the c -axis which is also the direction of the externally applied electric field E

$$\Delta n_3^2 = -\frac{1}{2} n_3^4 r_{33} E_3. \quad (4)$$

In this formula, n_3 is the component of the (diagonalized) dielectric tensor in the direction of the crystal axis and r is the pertinent electrooptic coefficient [6].

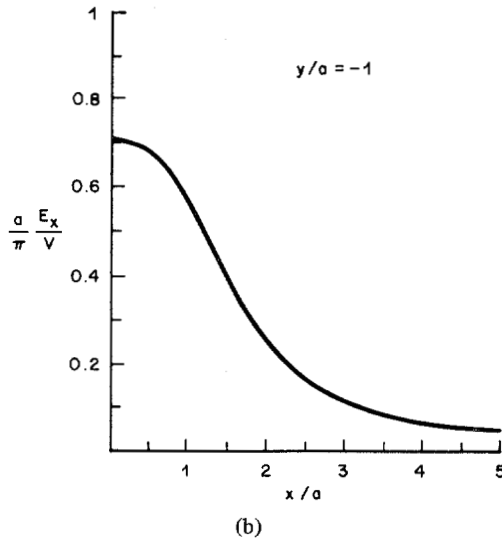
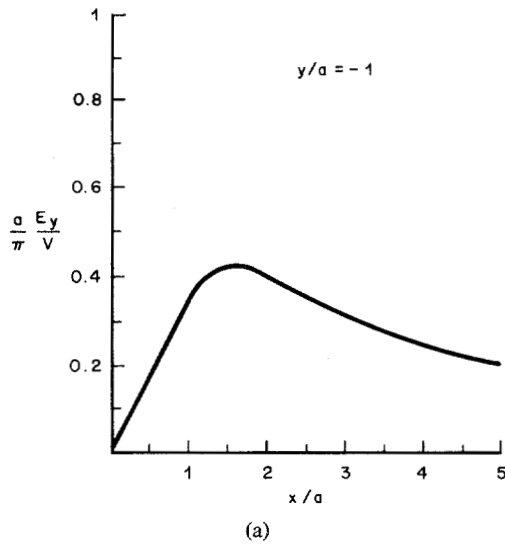
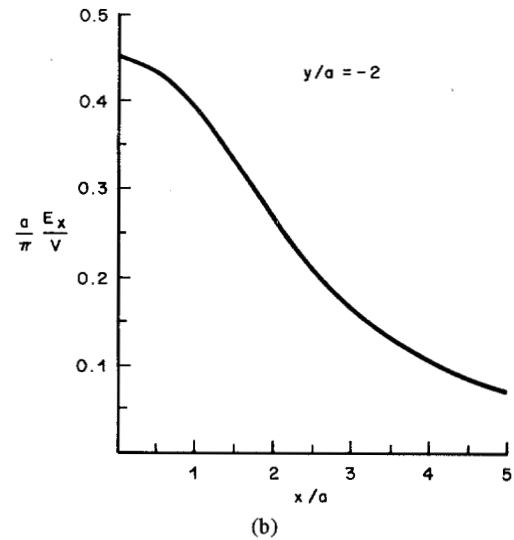
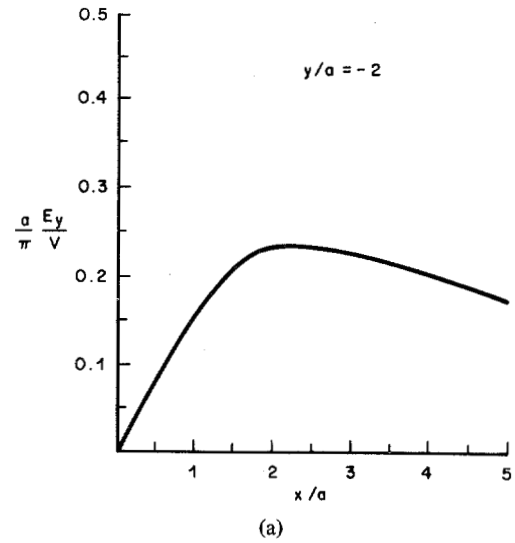
CHANGE OF THE PROPAGATION CONSTANT

The calculation of the change of the propagation constant in response to an externally applied signal voltage requires knowledge of the optical guided wave. An exact solution of the guided wave problem is hard to obtain. However, it is well established that the fields of the lowest order modes of weakly guiding dielectric optical waveguides have essentially only a single transverse electric field component that can be approximated by Hermite-Gaussian functions [7], [8]. Thus, we use the following approximation for the transverse component of the electric field vector of the guided mode field in the optical waveguide that is located beneath the metal electrodes

$$\mathcal{E} = \frac{2y}{w_x w_y^2 \sqrt{\pi}} \exp[-(y/w_y)^2] \exp\{-[(x-d)/w_x]^2\}. \quad (5)$$

In this formula d is the x position of the field maximum, w_x and w_y are width parameters that indicate how rapidly the field decays from its maximum value in x and y directions. While the x dependence of the field can be approximated by a Gaussian function, the y dependence of the field is slightly more complicated since the guided mode field is very weak at the surface of the crystal. Without the electrodes the field nearly vanishes at the surface (at $y = 0$) because of the large refractive index discontinuity. With electrodes, the waveguide structure is intentionally designed (by shaping its refractive index distribution) to ensure that the guided mode field is weak near the metal surface to minimize absorption losses. Thus, the field must be represented by a Hermite-Gaussian function. This explains the occurrence of the factor y in (5) which (except for a constant factor) is the Hermite polynomial of first order [9]. The function (5) is normalized so that the square of its integral over the infinite half space $0 < y < \infty$, $-\infty < x < \infty$ is equal to unity.

The approximation (5) of the guided mode must be used judiciously. It is adequate for describing the field in the region of high intensity and can thus be used for computing the change of the propagation constant. But it cannot be used for

Fig. 6. Same as Fig. 3 with $y/a = -1$.Fig. 7. Same as Fig. 3 with $y/a = -2$.

computing the coupling between the two guided modes of the directional coupler because the coupling coefficient depends on the weak evanescent tails of each field reaching into the region of the other, and the field tails are poorly approximated by (5).

The change $\Delta\beta$ of the propagation constant β of the guided modes can be expressed in terms of the change of the dielectric constant $\Delta(n_3^2)$ [as expressed by (4)] and the optical field $\mathcal{E}$ as [10]

$$\Delta\beta = \frac{k^2}{2\beta} \frac{\int_{-\infty}^0 dy \int_{-\infty}^{\infty} \Delta(n_3^2) |\mathcal{E}|^2 dx}{\int_{-\infty}^0 dy \int_{-\infty}^{\infty} |\mathcal{E}|^2 dx} \quad (6)$$

with the free space propagation constant $k = 2\pi/\lambda$.

Because of the symmetry of the problem, $\Delta(n_3^2)$ is equal in magnitude but has opposite signs for each of the two waveguides of the directional coupler. We are interested in maximizing the change of the propagation constant $\Delta\beta$ for fixed gap voltage V either as a function of waveguide position for

fixed gap width or as a function of the gap width for fixed position of the waveguides of the directional coupler.

OPTIMUM WAVEGUIDE POSITION OR OPTIMUM GAP WIDTH

The double integral in the denominator of (6) is unity because of the choice of normalization in (5). The double integral in the numerator with $\mathcal{E}$ given by (5) and $\Delta(n_3^2)$ given by (2) and (4) was evaluated numerically with the help of a computer. The results are plotted in arbitrary units (the same for all figures).

First, we consider the problem of positioning a single waveguide underneath a pair of electrodes of fixed gap width for the purpose of maximizing the induced change of the propagation constant (or phase shift). As an example, we assume that the mode field is wider in the x than in the y direction by choosing $w_y/w_x = 0.5$. Fig. 8 shows the relative phase shift as a function of the relative waveguide position d/w_x for several values of the normalized gap half width a/w_x . In this case it is assumed that the c -axis of the crystal is directed perpendicular to the surface so that the E_y component of the externally

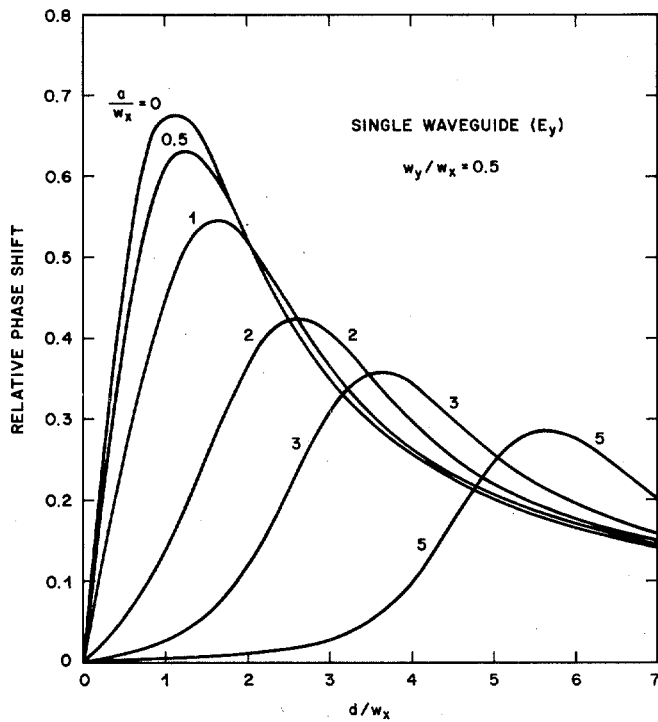


Fig. 8. Relative phase shift as function of normalized waveguide position d/w_x for several values of the normalized electrode half spacing a/w_x for the case that E_y is responsible for the phase shift.

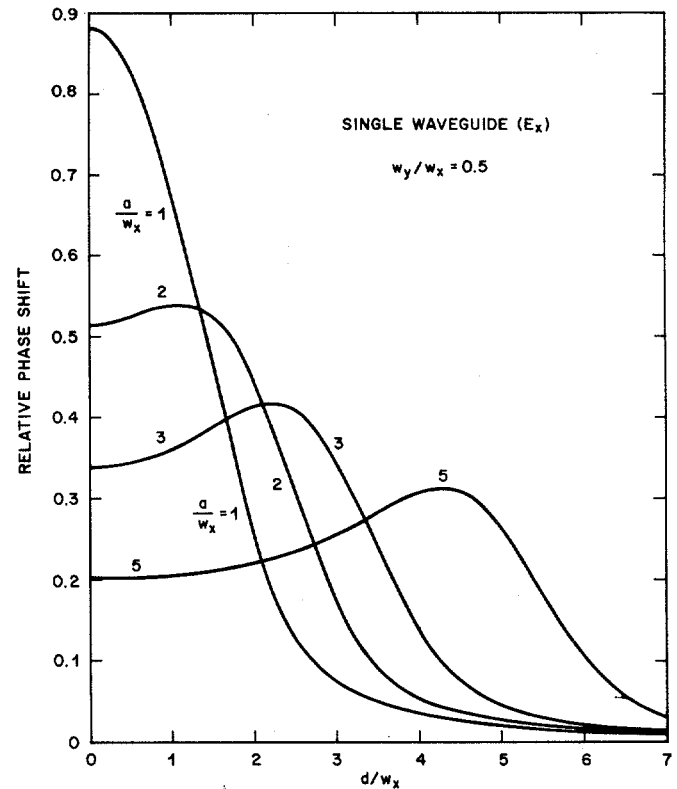


Fig. 9. Same as Fig. 8 except that now E_x is responsible for the phase shift.

applied electric field is responsible for the phase shift. Because the E_y field vanishes in the midplane between the electrodes, there is no change in propagation constant (no phase shift) at $d = 0$. The phase shift peaks when d is slightly larger than the gap half width $d > a$.

The situation changes when the crystal is cut so that the c -axis is parallel to the crystal surface and the E_x component of the applied field becomes responsible for the phase shift as shown in Fig. 9. Since E_x does not vanish at $x = 0$ the phase shift peaks at $d = 0$ for small values of the gap half width a , as shown by the curve labeled $a/w_x = 1$. For even smaller values of a the curve maintains its general shape but exhibits a higher peak that reaches the value 1.1 on the scale of Fig. 9. For larger gap width, $d = 0$ is no longer the optimum position. A slightly larger phase shift can be achieved by moving the waveguide into the vicinity of the edge of one electrode, but keeping $d < a$. The maxima of the curves in Figs. 8 and 9 decrease in height as a/w_x increases because for fixed voltage the field intensity decreases when the electrodes are moved farther apart.

Next, we consider the problem of finding an optimum electrode spacing $2a$ for maximizing the relative phase shift (change of propagation constants) of a push-pull directional coupler. Fig. 10 shows the relative phase shift as a function of the electrode gap width $2a$ for several values of the normalized waveguide position d/w_x . The curves in Fig. 10 were obtained by using $w_y/w_x = 1$ while the curves in Fig. 11 apply to mode fields whose extension in x direction is larger than their depth in y direction so that $w_y/w_x < 1$. The curves in Figs. 10 and 11 show that for small values of d/w the strongest "tuning" of the directional coupler occurs when the electrodes are spaced

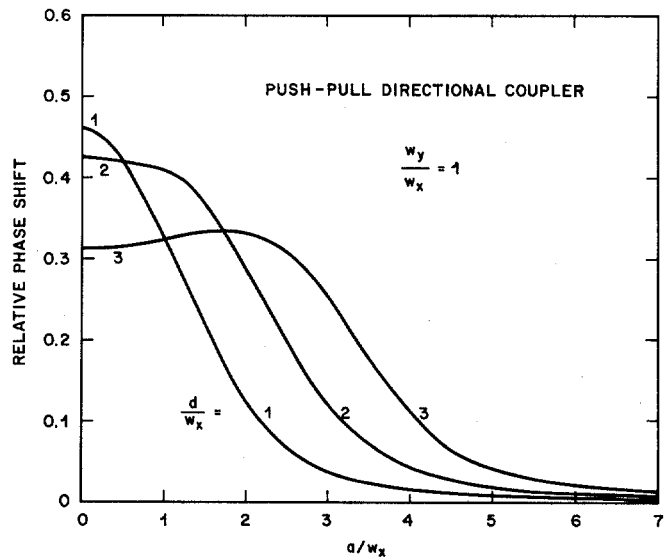


Fig. 10. Relative phase change of guided wave is shown as a function of the normalized gap half width a/w_x for fixed electrode voltage for several values of the normalized waveguide (half) separation $d/w_x \cdot a$ and for $w_y/w_x = 1$.

as close as possible $a = 0$. For mode fields whose depth is equal to their width ($w_y/w_x = 1$) a shallow maximum with $a/w_x > 0$ is observed only when the waveguides are moved sufficiently far apart ($d/w_x > 2$). For small waveguide separation, it is clearly best to space the electrodes as closely as possible.

Fig. 11 shows that the maximum of the change of the propagation constant moves away from $a = 0$ and becomes more

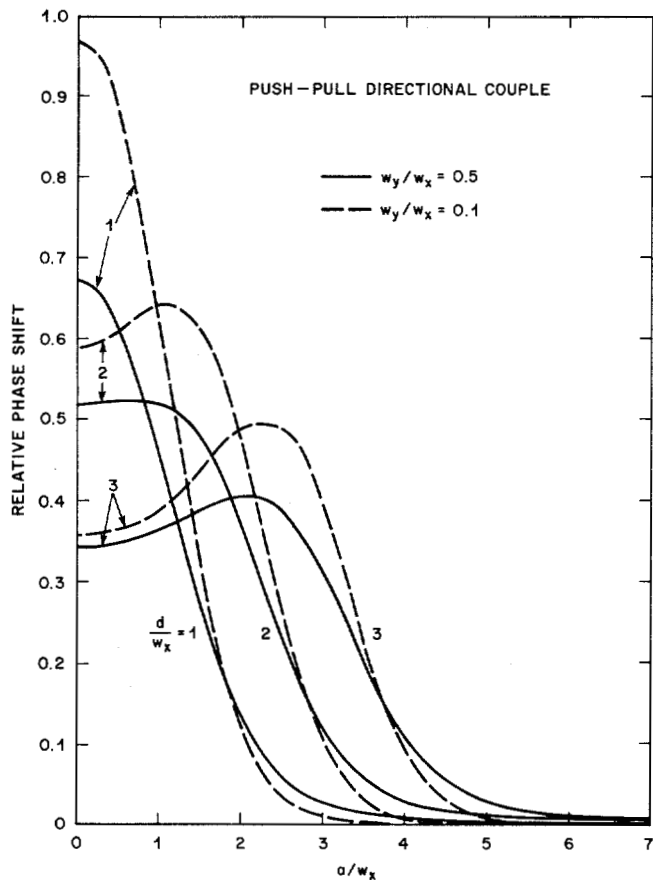


Fig. 11. Same as Fig. 10 for $w_y/w_x = 0.1$ and 0.5 .

pronounced as the mode fields become shallower. However, even in the case $d/w_x = 3$ and $w_y/w_x = 0.1$ the maximum value for an optimum electrode spacing $a/w_x = 2.25$ is only 36 percent larger than the change of the propagation constant that would be obtained at $a = 0$. For all other cases shown in Fig. 11 the advantage gained by using an optimum electrode spacing, rather than $a = 0$, is even less.

CONCLUSIONS

We have studied the influence of waveguide position and electrode spacing on the propagation constant of the guided modes of waveguides positioned underneath metal electrodes to which a constant potential is being applied. We found that the optimum position of a single waveguide depends on the component of the externally applied electric field that is effective in producing the desired change of the propagation constant of the guided mode. If the field component normal to the crystal surface is being used, the optimum waveguide position $x = d$ is near the edge $x = a$ of one of the electrodes with $d > a$. But if the field component parallel to the crystal surface is responsible for the desired change of the propagation constant, then the optimum waveguide spacing depends on the electrode separation. For closely spaced electrodes the wave-

guide should be placed beneath the center of the gap. For far spaced electrodes a slight advantage is to be gained by placing the waveguide near the edge of one of the electrodes with $d < a$.

For a push-pull directional coupler we found that in most cases of practical interest it is advantageous to space the electrodes as close to each other as possible. Only for very shallow guides that are spaced relatively far apart is a slight advantage to be gained by placing the edges of the electrodes nearer the centers of the waveguides. However, the optimum spacing does not occur when the edges of the electrodes are placed directly above the waveguide centers but closer to the common center line. Analytical expressions for optimum waveguide positions and optimum electrode spacings have so far not been found.

All linear dimensions occurring as variables or parameters in the curves shown in this paper are normalized with respect to the mode half width w_x . This means that the change of the propagation constant also appears as $w_x \Delta\beta$. So if w_x changes while the ratio of all other parameters relative to w_x are kept constant, $\Delta\beta$ would increase as $1/w_x$ for fixed gap voltage.

ACKNOWLEDGMENT

The author is grateful to R. C. Alferness for suggesting this problem and has benefited from several discussions with him.

REFERENCES

- [1] O. Mikami, J. Noda, and M. Fukuma, "Directional coupler type light modulator using LiNbO waveguides," *Trans. IECE Japan*, vol. E61, pp. 144-147, 1978.
- [2] K. Kubota, J. Noda, and O. Mikami, "Traveling wave optical modulator using a directional coupler LiNbO waveguide," *IEEE J. Quantum Electron.*, vol. QE-16, pp. 754-760, 1980.
- [3] P. S. Cross and R. V. Schmidt, "A 1 Gbit/second integrated optical modulator," *IEEE J. Quantum Electron.*, vol. QE-15, pp. 1415-1418, 1978.
- [4] R. C. Alferness, N. P. Economou, and L. L. Buhl, "Fast compact optical directional coupler switch/modulator," *Phys. Lett.*, vol. 38, pp. 214-216, 1981.
- [5] D. Marcuse, "Electrooptic coupling between the TE and TM modes in anisotropic slabs," *IEEE J. Quantum Electron.*, vol. QE-11, pp. 759-767, 1975.
- [6] I. P. Kaminow, *An Introduction to Electrooptic Devices*. New York: Academic, 1974.
- [7] D. Marcuse, "Gaussian approximation of the fundamental modes of graded-index fibers," *J. Opt. Soc. Amer.*, vol. 68, pp. 103-109, 1978.
- [8] A. W. Snyder, "Understanding monomode optical fibers," *Proc. IEEE*, vol. 69, pp. 6-13, 1981.
- [9] M. Abramowitz and I. A. Stegun, *Handbook of Mathematical Functions, Appl. Math. Series*, vol. 55. Washington, DC: Nat. Bureau of Standards, 1965.
- [10] J. A. Arnaud, *Beam and Fiber Optics*. New York: Academic, 1976, p. 217.

Dietrich Marcuse (M'58-F'73), for a photograph and biography, see p. 43 of the January 1982 issue of this JOURNAL.